

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1752

Greinwich 138 kV
June 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1752. The Interconnection Customer proposes a 37.9 MW solar photovoltaic facility interconnecting at the Greinwich 138 kV substation in the Pennsylvania area. The request was received on May 19, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 596 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,430,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Jack Crk to Talenenergy Martins Crk Llc W Shore 138.0 kV line
Contingency: Little Chute Substation to Talenenergy Martins Creek Llc Harrisburg Substation 345.0 kV
Loading: 139.6% (Rating: 428 MVA)
DFAX: 6.1%
Pre-project loading: 139.1%
2. Facility: Little Chute 345/138kV TX
Contingency: Little Chute to Talenenergy Martins Crk Llc Harrisburg 345.0 kV
Loading: 130.3% (Rating: 382 MVA)
DFAX: 5.7%
Pre-project loading: 129.7%
3. Facility: Mountain to Greinwich 138.0 kV
Contingency: Mountan 345 kV/138 kV Autotransformer
Loading: 117.8% (Rating: 227 MVA)
DFAX: 27.1%
Pre-project loading: 113.2%
4. Facility: Mcclelln-Adc to Deerfield 5 138.0 kV
Contingency: Little Chute to Talenenergy Martins Creek Llc Harrisburg 345.0 kV
Loading: 116.5% (Rating: 260 MVA)
DFAX: 8.6%
Pre-project loading: 115.3%
5. Facility: Elk City Llc to John Hart 138.0 kV line
Contingency: Little Chute to Talenenergy Martis Crk Llc Harrisburg 345.0 kV line
Loading: 114.3% (Rating: 293 MVA)
DFAX: 5.7%
Pre-project loading: 113.5%

6. Facility: Calaveras to Panguitch 138.0 kV
Contingency: Elk City Llc to Little Chute Substation 345.0 kV
Loading: 107.4% (Rating: 228 MVA)
DFAX: 6.6%
Pre-project loading: 106.3%
7. Facility: Pingstn to Calaveras 138.0 kV Line
Contingency: Elk City Llc to Little Chute Substation 345.0 kV
Loading: 107.2% (Rating: 204 MVA)
DFAX: 6.6%
Pre-project loading: 105.9%
8. Facility: Hunterstown Power Plant to Deerfield 5 138.0 kV L
Contingency: Hastings to Elk City Llc 345.0 kV
Loading: 101.2% (Rating: 386 MVA)
DFAX: 21.4%
Pre-project loading: 99.1%
9. Facility: Sawtelle to Hunterstown Power Plant 138.0 kV
Contingency: Hastings to Elk City Llc Substation 345.0 kV line
Loading: 100.7% (Rating: 406 MVA)
DFAX: 21.4%
Pre-project loading: 98.7%
10. Facility: Sawtelle to Hood Riv 138.0 kV L
Contingency: Hastings to Elk City Llc 345.0 kV L
Loading: 100.5% (Rating: 415 MVA)
DFAX: 21.4%
Pre-project loading: 98.6%
11. Facility: Rainey to Hood River 138.0 kV
Contingency: Hastings to Elk City Llc 345.0 kV line
Loading: 100.1% (Rating: 436 MVA)
DFAX: 21.4%
Pre-project loading: 98.2%
12. Facility: Elk City Llc to Little Chute 345.0 kV
Contingency: Fort Humbug to Little Chute 345.0 kV
Loading: 100.0% (Rating: 1,135 MVA)
DFAX: 24.6%
Pre-project loading: 99.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Talenenergy Martins Crk Llc W Shore to Jack Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$2,189,000	16% (\$345,000)
Little Chute 345-138 kV TX	Replace transformer with higher-rated unit	\$23,425,000	5% (\$1,231,000)
Greinwich to Mtn 138.0 kV	Rebuild line with higher-capacity conductor	\$3,142,000	100% (\$3,142,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcclellan-Adc to Deerfield 5 Substation 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,800,000	100% (\$3,800,000)
Elk City Llc to John Hart 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,204,000	11% (\$136,000)
Calaveras to Panguitch 138.0 kV	Reconductor line	\$778,000	100% (\$778,000)
Pingstn to Calaveras 138.0 kV Line	Reconductor line	\$2,877,000	100% (\$2,877,000)
Deerfield 5 to Hunterstown Power Plant 138.0 kV	Reconductor line	\$1,302,000	41% (\$536,000)
Sawtelle to Hunterstown Power Plant 138.0 kV	Reconductor line	\$2,296,000	41% (\$946,000)
Sawtelle to Hood Riv 138.0 kV	Reconductor line	\$1,636,000	41% (\$674,000)
Rainey to Hood River 138.0 kV	Reconductor line	\$1,206,000	41% (\$497,000)
Little Chute to Elk City Llc 345.0 kV	Reconductor line	\$16,566,000	3% (\$468,000)
	Total allocated to project		\$15,430,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.